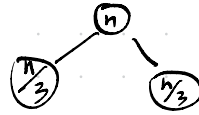


6. Recurrence Relations

(a) $T(n) = 2T(n/3) + O(n)$

递推: $2^{\log_3 n} = n^{\log_3 2}$

$n(1 + \frac{2}{3} + \frac{2^2}{3^2} + \dots) \in O(n)$



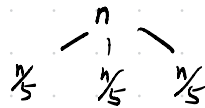
n
 $\frac{n}{3} + \frac{n}{3}$

$2 < 3^1$

$a < b^d$

(b) $T(n) = 3T(n/5) + O(n \log n)$

Recursive: $3^{\log_5 n} = n^{\log_5 3}$



$n \log n$

$\sum_{k=0}^{\log_5 n} n \left(\frac{3}{5}\right)^k \log \frac{n}{5^k}$

$\log n - k \log 5$

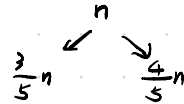
$\in O(n \log n)$

$S = 1 + \frac{3}{5} \cdot 1 + \left(\frac{3}{5}\right)^2 \cdot 2 + \dots$

$\frac{3}{5}S = \frac{3}{5} + \left(\frac{3}{5}\right)^2 \cdot 1 + \left(\frac{3}{5}\right)^3 \cdot 2$

$\frac{2}{5}S = 1 + \left(\frac{3}{5}\right)^2 + \dots$

(c) $T(n) = T\left(\frac{3n}{5}\right) + T\left(\frac{4n}{5}\right)$



k-Layer: $\left(\frac{3}{5}\right)^i \left(\frac{4}{5}\right)^j n$ $i+j=k$ $\left(\frac{3}{5}\right)^2 \left(\frac{4}{5}\right)^1 n$ $\left(\frac{3}{5}\right)^1 \left(\frac{4}{5}\right)^2 n$ $\left(\frac{4}{5}\right)^3 n$

叶子有几个: $\left(\frac{3}{5}\right)^i \left(\frac{4}{5}\right)^j n = 1$

$\log_{\frac{7}{5}} n \rightarrow \log\left(\frac{7}{5}\right) n$ $O(\ln n)$

$$(d) \quad T(n) = T(\sqrt{n}) + 1$$

$$n^{\frac{1}{2^k}} = 1 \quad \log_n 2 = \frac{1}{2^k}$$

$$\log_2(\log_2 n)$$

$$O(\ln(\ln n))$$

$$\sqrt[n]{n} \leftarrow \sqrt[n]{n} \leftarrow n$$

$$\frac{1}{1}$$